

Objective

To implement and demonstrate Constructor Inheritance and Method Inheritance in Java, where class Y extends class X and uses the superclass constructor, variable, and method.

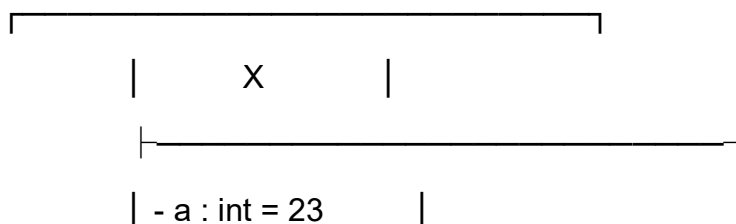
Steps for Execution

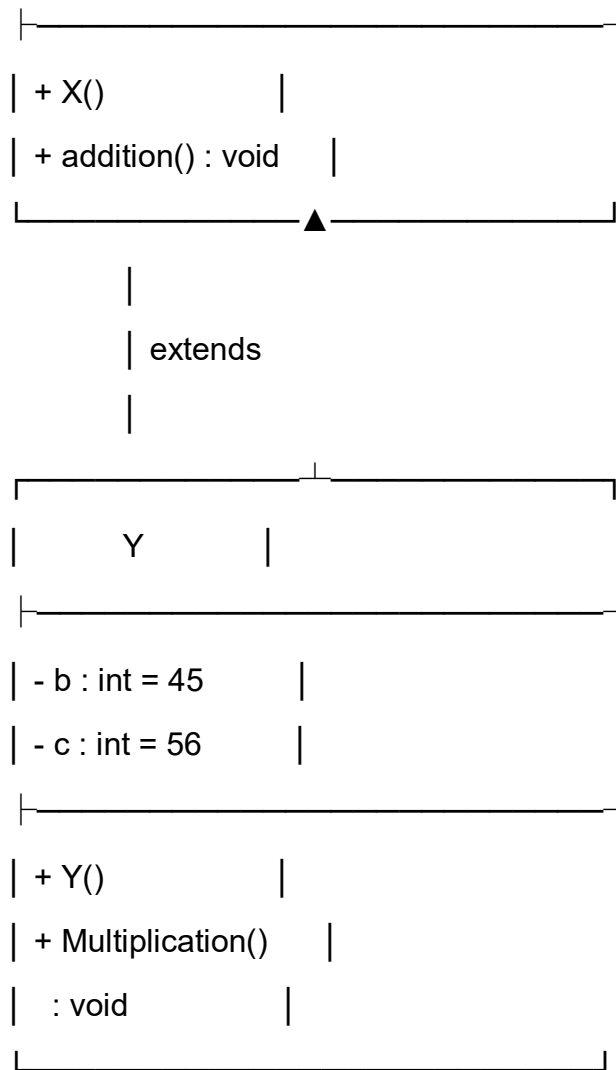
1. Create a Java project and package named javacore.
2. Create a file named ClassDemo.java and enter the given code.
3. Define the superclass X with a constructor and addition() method.
4. Define the subclass Y extending X with a constructor and Multiplication() method.
5. Create an object of Y in the main() method.
6. Compile and run the program.
7. Observe the output.

Algorithm

1. Start the program.
2. Define class X with variable a, constructor, and addition().
3. Define class Y extending X with variables b, c, constructor, and Multiplication().
4. Create an object of Y.
5. Execute the superclass constructor.
6. Call addition() and Multiplication().
7. Display the results.
8. Stop the program.

UML CLASS DIAGRAM:





Relationship

- Y extends X
- Y inherits the a variable and addition() method from X.
- When a Y object is created, the X constructor executes first, followed by the Y constructor.

CODE:

```

package javacore;

class X{
    int a=23;

```

```
public X(){
System.out.println("this is a superclass constructor");
}

void addition(){
    System.out.println("add is "+(a+2350));
}
}

class Y extends X{
    int b=45,c=56;

    public Y(){
        System.out.println("this is subclass constructor");
    }

    void Multiplication(){
        System.out.println("multiplication of b and c is "+(b*c));
    }
}

public class ClassDemo {
    public static void main(String[] args){
        Y ac=new Y();
        ac.addition();
        ac.Multiplication();
    }
}
```

OUTPUT:

this is a superclass constructor

this is subclass constructor

add is 2373

multiplication of b and c is 2520.

Result:

The program successfully demonstrates Inheritance with Constructors and Methods in Java. When an object of subclass Y is created, the superclass X constructor executes first, followed by the subclass Y constructor. The inherited addition() method and the subclass Multiplication() method are then executed successfully